

VBASE3: Must Read Before Using

⚠ COMMUNITY ADVISORY & BETA NOTICE

VBASE3 is currently in beta phase. It is generated by the interaction of one person and AI. It most likely contains errors that are difficult to detect or even stupid simple ones.

After feedback with the scientific community, the database and the software will get more robust. After the initial release, the harvester will start again and the database content will be further improved.

✉ **Feedback & Error Submissions:** Please email any corrections or comments to:
Prof. Werner Müller (werner.muller@manchester.ac.uk), Emeritus Professor of Cellular Immunology, University of Manchester.

Scope of the Beta Release Catalogs

Class I Confirmed Functional Catalogs

2,856 verified genes across 36 benchmark taxa. Every sequence is validated by public rearranged repertoire alignments ($\geq 75\%$ identity across mature FR1 $\rightarrow$ Cys104).

Synthesis Boundary Compliance

All V-genes terminate strictly at conserved Cys104. Heavy J-segments end at **TVSS / TVSA** ; Light J-segments end at **TVL / TAL** or **VEIK / LEIK / VDIK** with zero constant region bleed-through.

TCRBase Pan-Vertebrate Germlines

3,114 curated TCR genes covering TRA, TRB, TRG, and TRD loci across 31 vertebrate species with reading-frame and CDR3 anchor validation.

Rosetta Stone & Constant Cassettes

3,900 cross-references uniting IMGT, VBASE2, and Tomlinson DP nomenclature, plus 108 modular constant cassettes with 100% INSDC GenBank traceability.

Client-Side Web Portals (Zero-Backend WebAssembly)

All 26 interactive web visualizers execute 100% client-side in your web browser using compiled Rust WebAssembly (`dnaplot_wasm_bg.wasm`). Zero sequences, clone tables, or queries are ever sent to external servers. Hosted on GitHub Pages: https://vbase3.github.io/vbase3_database/.

How to Report Discrepancies

- **Sequence or Allele Anomalies:** If an alignment shows an unexpected frameshift, boundary overhang, or missing functional allele with published empirical backing.
- **Nomenclature Mappings:** If a cross-reference between Tomlinson, IMGT, or VBASE2 requires refinement.
- **Engine Edge Cases:** If the WebAssembly sequence analyzer produces unexpected boundary detections.

VBASE3 Initiative • University of
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GitHub:

github.com/vbase3/vbase3_database

Archival Deposit:

doi.org/10.5281/zenodo.22923345